

Question 13

Q: Explain how the features of Computer Vision systems enable applications such as automated surveillance.

Theoretical Concept: CV Features in Surveillance

Automated surveillance relies on specific Computer Vision features:

- **Optical Flow:** Tracking movement direction and speed.
- **Background Subtraction:** Isolating moving intruders from static environments.
- **Object Detection/Tracking:** Drawing bounding boxes around humans or vehicles (YOLO/SSD).

OpenCV Python Solution

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```
import cv2 # Import OpenCV for image processing algorithms
import numpy as np # Import Numpy for mathematical matrix operations
import matplotlib.pyplot as plt # Import Matplotlib to visualize the output on screen

# Step 1: Create two sequential frames simulating an object moving to the right
frame1 = np.zeros((100, 100), dtype=np.uint8) # Create a blank black image array filled with zeros
cv2.rectangle(frame1, (20, 20), (40, 40), 255, -1) # Draw a filled or outlined rectangle on the image
```

```
frame2 = np.zeros((100, 100), dtype=np.uint8) # Create a blank black image array filled with zeros
cv2.rectangle(frame2, (40, 20), (60, 40), 255, -1) # Object moved +20 pixels

# Step 2: Compute Dense Optical Flow (A core surveillance algorithm)
# This calculates the velocity vector of every pixel between the two frames
flow = cv2.calcOpticalFlowFarneback(frame1, frame2, None, 0.5, 3, 15, 3, 5, 1.2, 0) # Compute dense optical flow b
mag, ang = cv2.cartToPolar(flow[..., 0], flow[..., 1]) # Convert Cartesian (x,y) vectors to polar (mag, angle)

# Step 3: Visualize the motion magnitude
plt.imshow(mag, cmap='hot') # Render the image array to the display
plt.title('Optical Flow Magnitude (Motion Tracking)') # Add a descriptive title to the plot
plt.colorbar() # Add a color legend bar to the plot
plt.show() # Show the final plot window to the user
```

Expected Output

The output plot shows a 'heat map' (hot colormap) where the exact region that the object traveled through glows brightly, proving the algorithm tracked its motion vectors successfully.

How to Execute & Submit

1. Google Colab Execution:

- Open [Google Colab](#) and create a New Notebook.
- Click the  **Copy** button on the code block above.

- Paste it into a Colab cell and press **Shift + Enter** to run.

2. Uploading to GitHub:

- In Colab, go to **File > Save a copy in GitHub**.
- Authorize your GitHub account.
- Select your Computer Vision Lab repository and click **OK** to commit the code.